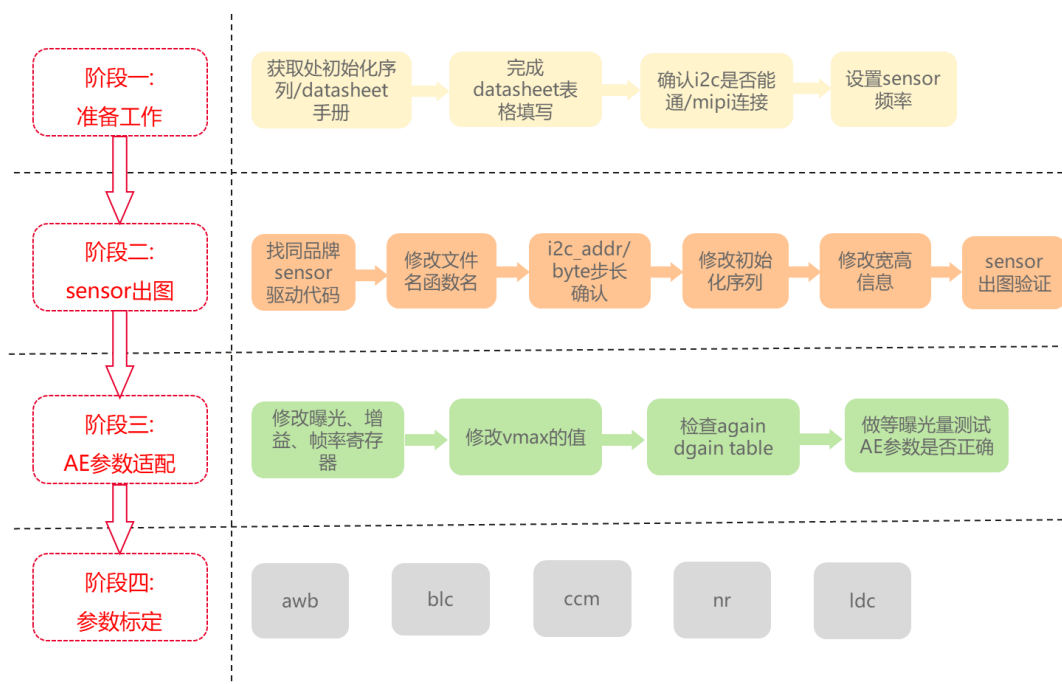


前言



适配sensor出图分为如下四个阶段：

阶段一：准备工作。获取sensor初始化序列、datasheet表格、确认i2c/mipi硬件连接、sensor时钟配置。

阶段二：sensor出图。也是最重要的环节，当mipi_rx中宽高正确，vi中有帧率，此时即可开始mpp模块的开发。

阶段三：AE参数配置。需要完成帧率配置、曝光时间、增益相关寄存器和函数的适配。修改帧率是否能达到最大值和最小值；曝光是否能达到最大值和最小值；增益是否能达到最大值和最小值；等曝光量测试是否能通过。

阶段四：参数标定。sensor驱动代码cmos_param.h/cmos_ex.h，AWB (自动白平衡)、BLC (背光补偿)、CCM (色彩校正矩阵)、NR (降噪)、LDC (镜头畸变校正)。

1 准备工作

1.1 完成 datasheet 表格的寄存器信息、

找sensor原厂fae获取sensor datasheet数据手册文档，完成datasheet表格中寄存器地址的填写。

1.2 sensor 初始化序列获取

找sensor原厂fae获取符合海思芯片的初始化序列。

1.3 确认硬件是否连通

确认i2c硬件能读写，设置i2c的sda scl管脚复用， sensor_crg, sensor0_scl、sensor0_rst的管脚复用设置，以Hi3516CV610为例

```
bspmm 0x17940098 0x1135 #i2c_sda
```

```
bspmm 0x1794009c 0x1135 #i2c_scl
```

```
bspmm 0x11018440 0xa010 #sensor_crg
```

```
bspmm 0x17940084 0x1211 #sensor0_scl
```

```
bspmm 0x1794008c 0x1135 #sensor0_rst
```

i2c读取数据，不会报错。

例如:

```
i2c_read 0x0 0x60 0x3401 0x3405 2 1
```

0x0表示i2c设备号

0x60表示i2c地址

0x3401、0x3405表示sensor中的寄存器地址，需要依次读取0x3401-0x3405 5个寄存器的值，

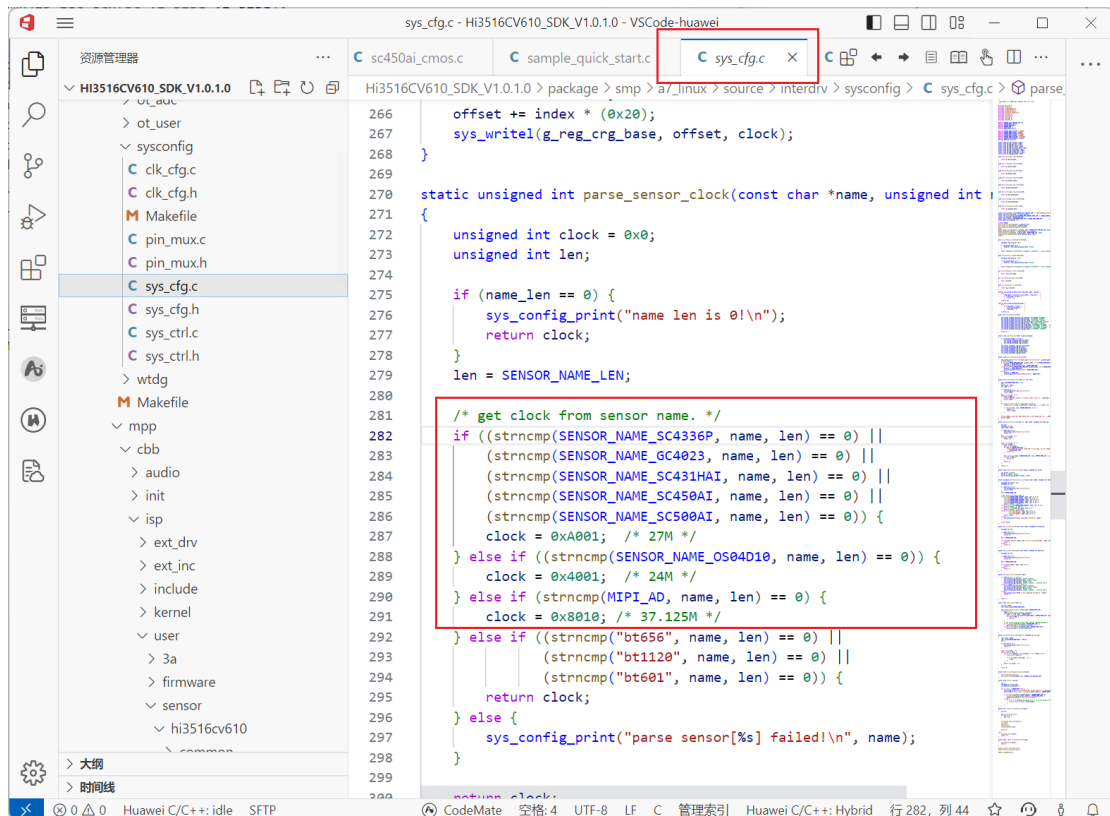
2 1 表示步长参数

如sensor的寄存器i2c_read 0x0 0x60 0x34 0x34 **1** 1，步长也需要修改。

1.4 sensor 频率设置

如果sensor的时钟是24M 或者27M，需要在sys_cfg.c parse_sensor_clock() 函数中添加自己的sensor名字，来设置频率。随后重新编译，生成新的sys_config.ko，替换一下新的ko文件。

在加载ko时的命令，例如./load3516cv610_20s_debug -a -sensor0 sc4336p 这里的sensor0 sc4336p参数就是从这里解析的，然后设置的输出频率。

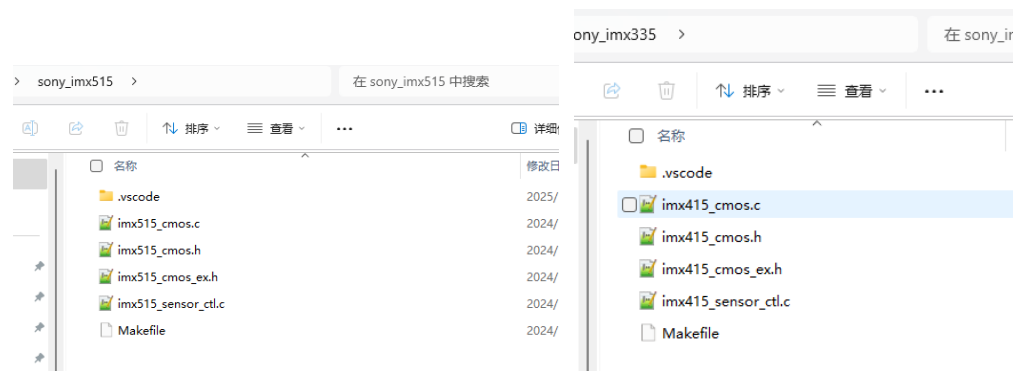


2 Sensor 驱动适配出图

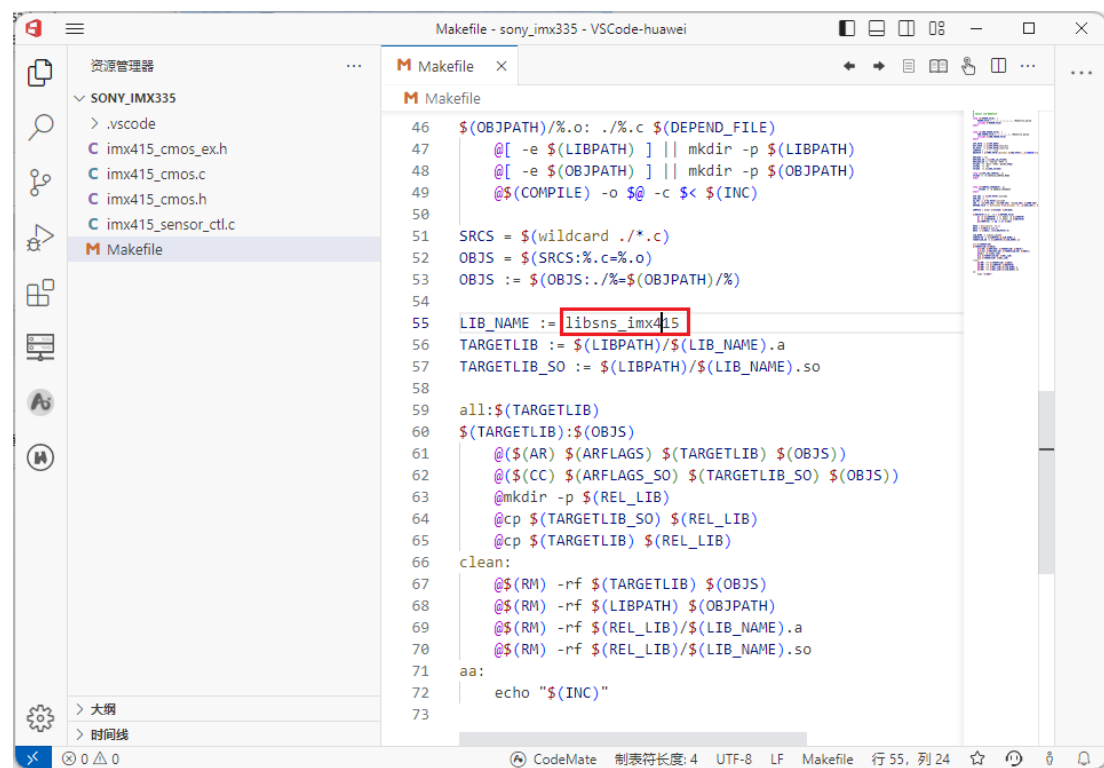
详细步骤参考《sensor调试指南》文档

本章举例，需要适配sony_imx415 线性模式 驱动，在sdk包中找到了imx515 线性模式的驱动，所以在imx515驱动代码的基础上进行适配，其他品牌用同样的方法找到同品牌的sensor驱动进行适配。下面介绍手动适配的步骤。

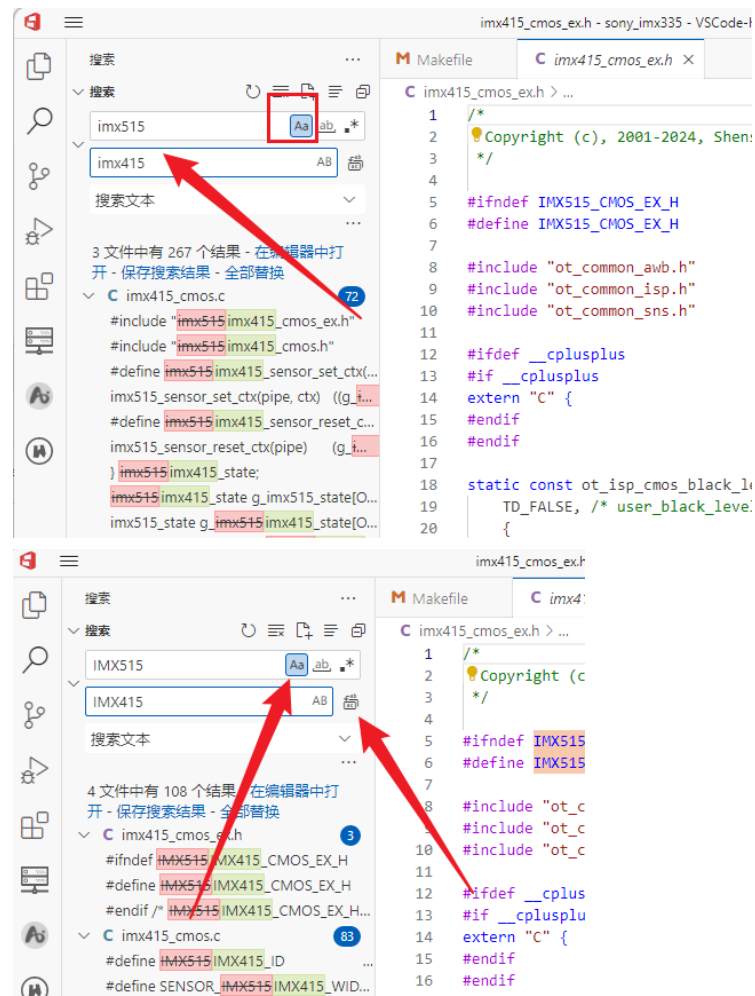
2.1 修改 cmos.c .h 文件名



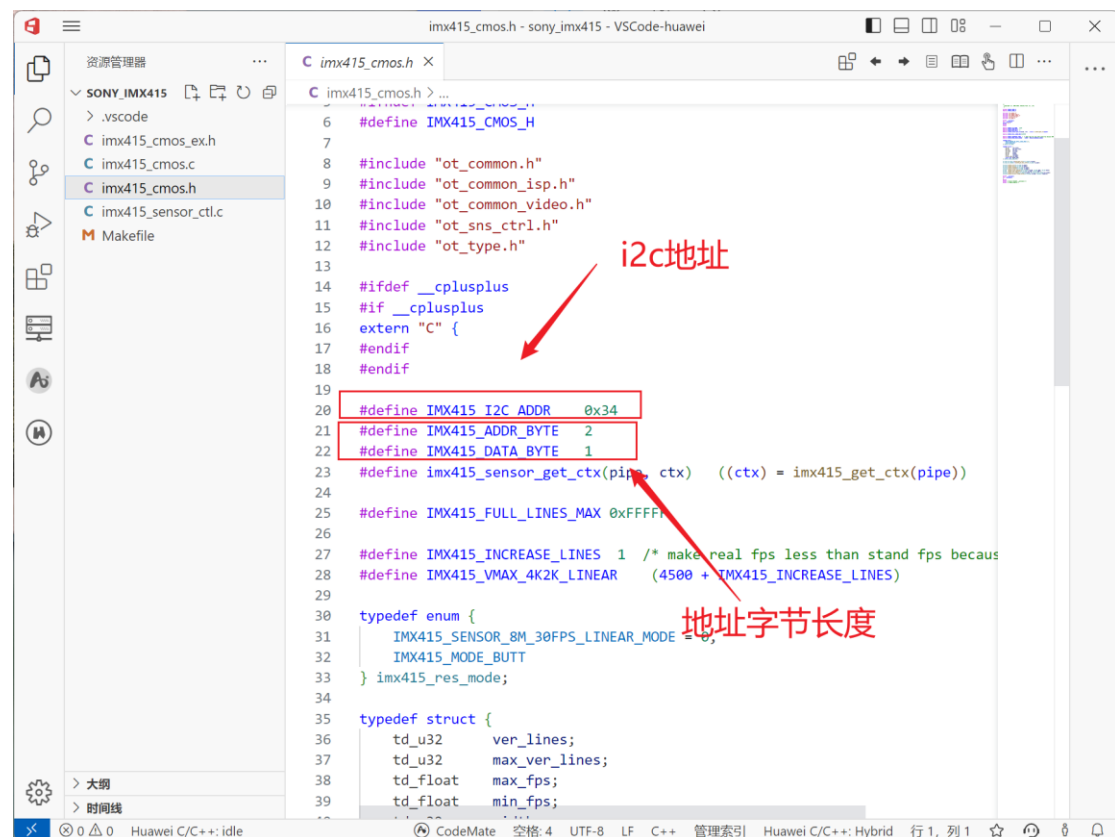
2.2 修改 makefile 中的 lib_name



2.3 全局替换函数名（注意函数名称区分大小写）



2.4 修改 i2c addr



检查sensor的i2c地址

检查地址长度，

例如sensor的某个寄存器的地址是0x36D8

```
#define IMX415_ADDR_BYTE 2
```

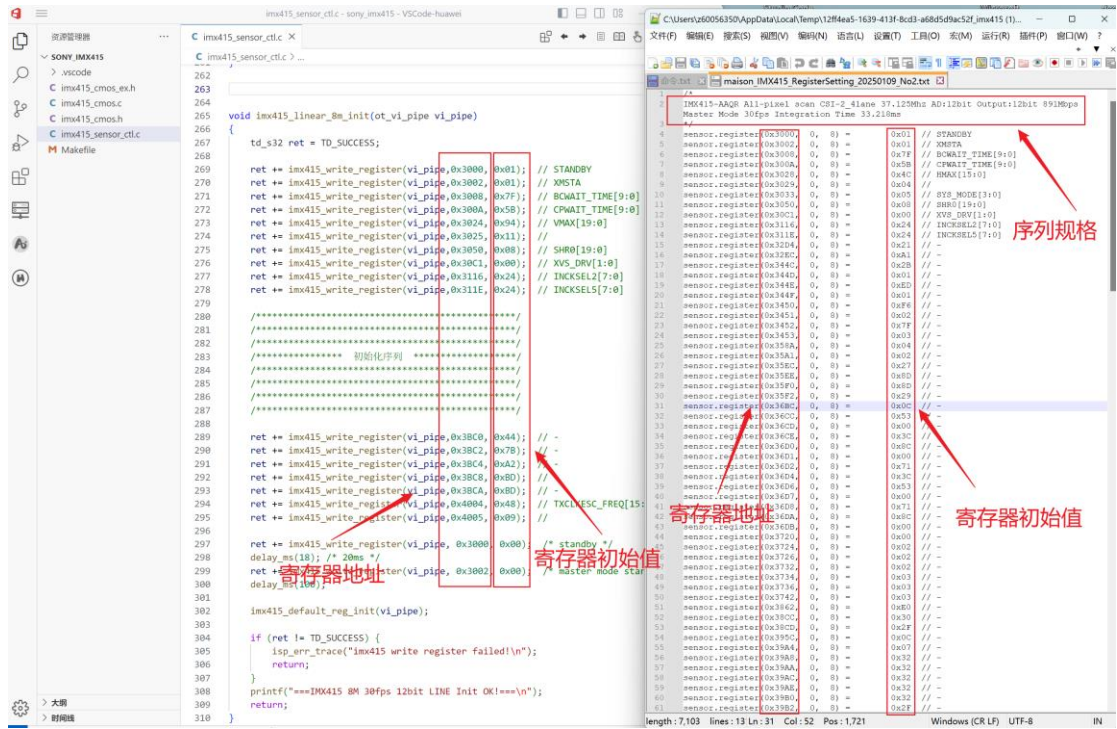
```
#define IMX415_DATA_BYTE 1
```

再例如sensor的某个寄存器地址是0x5e

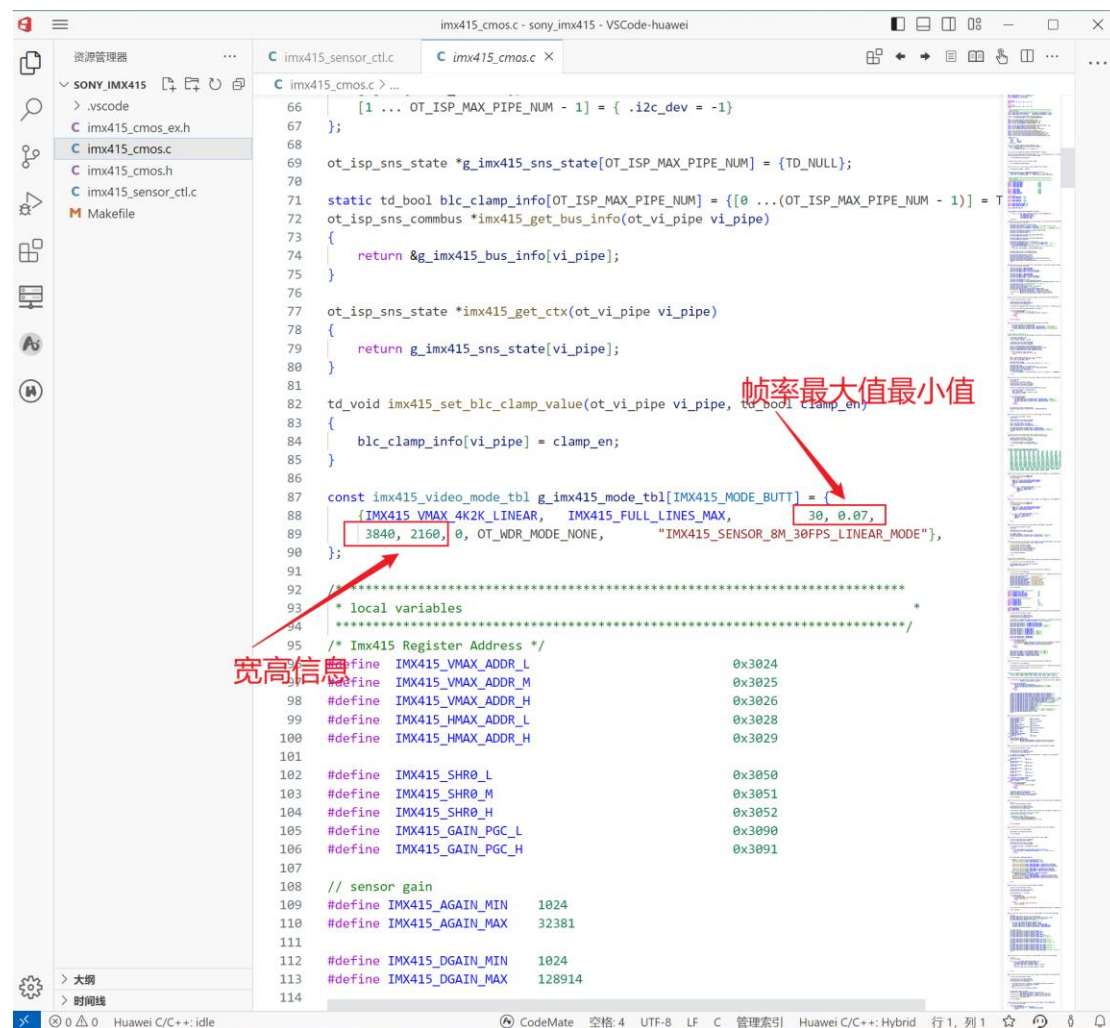
```
#define IMX415_ADDR_BYTE 1 这里需要注意
```

```
#define IMX415_DATA_BYTE 1
```

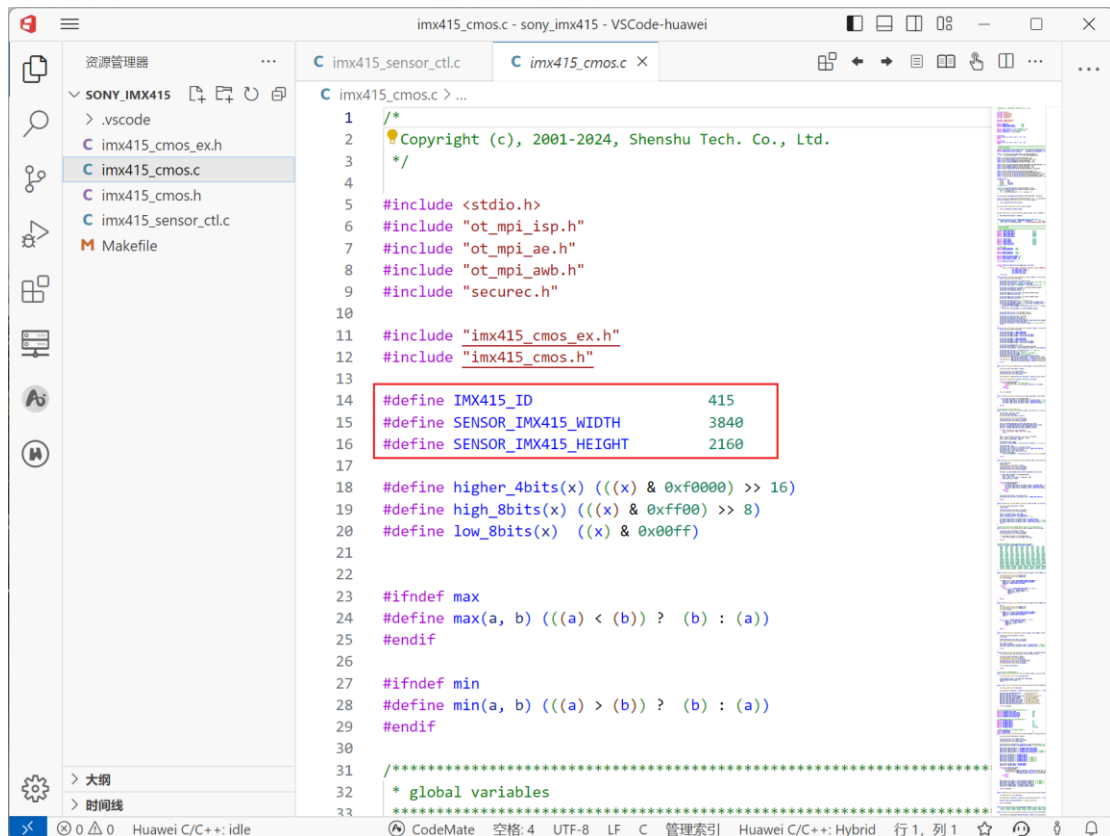
2.5 修改初始化序列代码



2.6 修改宽高信息，帧率信息

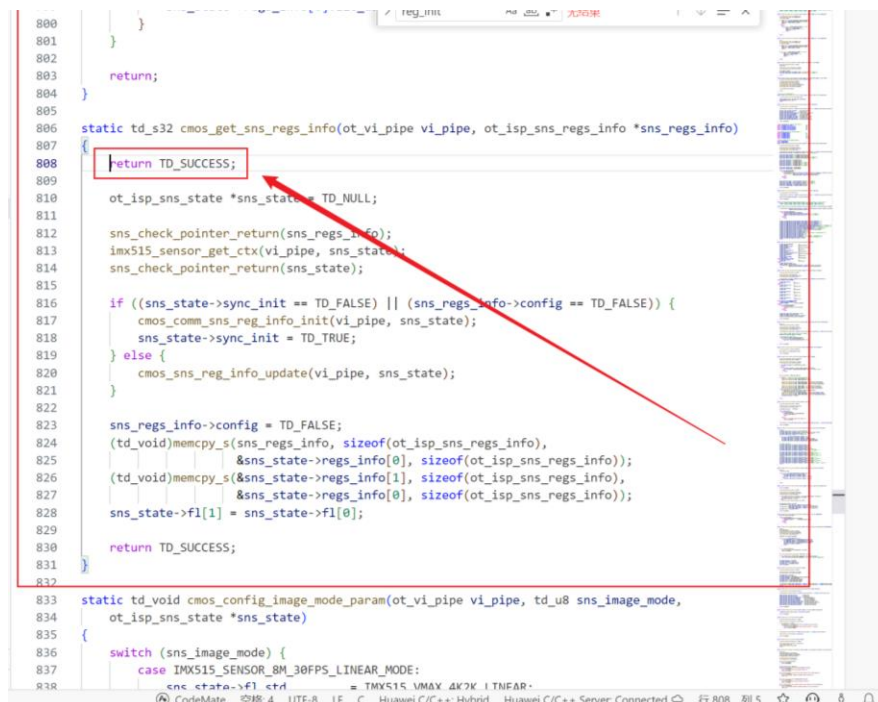


```
66 [1 ... OT_ISP_MAX_PIPE_NUM - 1] = { .i2c_dev = -1 }
67 };
68
69 ot_isp_sns_state *g_imx415_sns_state[OT_ISP_MAX_PIPE_NUM] = {TD_NULL};
70
71 static td_bool blc_clamp_info[OT_ISP_MAX_PIPE_NUM] = {[0 ... (OT_ISP_MAX_PIPE_NUM - 1)] = T
72 ot_isp_sns_commbus *imx415_get_bus_info(ot_vi_pipe vi_pipe)
73 {
74     return &g_imx415_bus_info[vi_pipe];
75 }
76
77 ot_isp_sns_state *imx415_get_ctx(ot_vi_pipe vi_pipe)
78 {
79     return g_imx415_sns_state[vi_pipe];
80 }
81
82 td_void imx415_set_blc_clamp_value(ot_vi_pipe vi_pipe, td_bool clamp_en)
83 {
84     blc_clamp_info[vi_pipe] = clamp_en;
85 }
86
87 const imx415_video_mode_tbl g_imx415_mode_tbl[IMX415_MODE_BUTT] = {
88     {IMX415_VMAX_4K2K_LINEAR, IMX415_FULL_LINES_MAX, 30, 0.07,
89      3840, 2160, 0, OT_WDR_MODE_NONE, "IMX415_SENSOR_8M_30FPS_LINEAR_MODE"},
90 };
91
92 /* *****
93  * local variables
94  * *****
95
96 /* Imx415 Register Address */
97 #define IMX415_VMAX_ADDR_L 0x3024
98 #define IMX415_VMAX_ADDR_M 0x3025
99 #define IMX415_VMAX_ADDR_H 0x3026
100 #define IMX415_HMAX_ADDR_L 0x3028
101 #define IMX415_HMAX_ADDR_H 0x3029
102
103 #define IMX415_SHR0_L 0x3050
104 #define IMX415_SHR0_M 0x3051
105 #define IMX415_SHR0_H 0x3052
106 #define IMX415_GAIN_PGC_L 0x3090
107 #define IMX415_GAIN_PGC_H 0x3091
108
109 // sensor gain
110 #define IMX415_AGAIN_MIN 1024
111 #define IMX415_AGAIN_MAX 32381
112
113 #define IMX415_DGAIN_MIN 1024
114 #define IMX415_DGAIN_MAX 128914
```



2.7 验证 sensor 出图

验证sensor出图时，可以认为仅跑初初始化序列，可以直接在cmos.c文件中的cmos_get_sns_regs_info()函数入口处，直接return TD_SUCCESS。相当于不使用isp中的ae参数，也就是使用初始化序列中的默认曝光、增益、帧率配置。



验证sensor出图时，可以cat /proc/umap/mipi_rx，查看是否出宽高信息。

```
/mnt/610/Hi3516CV610_P0_V1.0.1.0 # cat /proc/umap/mipi_rx
Module: [MIPI_RX], Build Time[Sep 7 2024, 17:51:12]

-----lane_mode-----
lane_mode      :4          |
-----port_attr-----
port_id        :0          |work_mode      :mipi    |data_rate      :x1      |data_type      :raw_12
img_rect_x     :0          |img_rect_y     :0          |img_width      :2560     |img_height     :1440
lane_id        :0,2,-1,-1 |wdr_mode       :none     |ext_dt_1       :0x2c     |ext_dt_width_1 :12
ext_dt_2       :0x2c     |ext_dt_width_2 :12      |ext_dt_3       :0x2c     |ext_dt_width_3 :12
-----clk_status-----
sensor_clk     :Y          |mipi_clk       :Y          |
-----phy_info-----
phy_id         :0          |phy_en         :Y          |
phy_d0_term_en :Y          |phy_d1_term_en :N          |phy_d2_term_en :Y          |phy_d3_term_en :N
lane0_en       :Y          |lane1_en       :N          |lane2_en       :Y          |lane3_en       :N
lane0_map      :0x0       |lane1_map      :0x1       |lane2_map      :0x1       |lane3_map      :0x3
cil_d0_cur_stat :HS_LP00  |cil_d0_nxt_stat :HS_LP00  |cil_d1_cur_stat :IDLE     |cil_d1_nxt_stat :IDLE
cil_d2_cur_stat :HS_LP00  |cil_d2_nxt_stat :HS_LP00  |cil_d3_cur_stat :IDLE     |cil_d3_nxt_stat :IDLE
cil_clk_cur_stat :HS_LP00  |cil_clk_nxt_stat :HS_LP00  |cil_clk2_cur_stat :IDLE   |cil_clk2_nxt_stat :IDLE
-----data_detect-----
phy_id         :0          |freq_measure    :632MHz   |lane2_data      :0xbc     |lane3_data      :0x0
lane0_data     :0x64     |lane1_data      :0x0       |mipi_ph_d2      :0x80     |mipi_ph_d3      :0x0
mipi_ph_d0     :0x2b     |mipi_ph_d1      :0x0       |mipi_data2      :0x2d     |mipi_data3      :0x0
mipi_data0     :0x27     |mipi_data1      :0x0       |mipi_vc1_w      :0          |mipi_vc1_h      :0
mipi_vc0_w     :2560     |mipi_vc0_h      :1440     |mipi_vc3_w      :0          |mipi_vc3_h      :0
-----phy_err-----
phy_id         :0          |clk_fsm_timeout :0          |d0_fsm_timeout  :0          |d1_fsm_timeout  :0
d2_fsm_timeout :0          |d3_fsm_timeout  :0          |d0_err_escape   :1          |d1_err_escape   :0
d2_err_escape  :0          |d3_err_escape   :0          |
-----mipi_err-----
vc0_crc_err    :0          |vc1_crc_err     :0          |vc2_crc_err     :0          |vc3_crc_err     :0
vc0_ecc_err    :0          |vc1_ecc_err     :0          |vc2_ecc_err     :0          |vc3_ecc_err     :0
vc0_frm_num_err :0          |vc1_frm_num_err :0          |vc2_frm_num_err :0          |vc3_frm_num_err :0
vc0_frm_mismatch :0        |vc1_frm_mismatch :0        |vc2_frm_mismatch :0        |vc3_frm_mismatch :0
vsync_cnt      :10         |rd_data_fifo_err :0          |wr_data_fifo_err :0          |rd_cmd_fifo_err :0
wr_cmd_fifo_err :0          |data_fifo_n_empt :0          |cmd_fifo_n_empt :0          |
-----align_err-----
fifo_ovfl      :0          |lane0_fifo_ovfl :0          |lane1_fifo_ovfl :0          |lane2_fifo_ovfl :0
lane3_fifo_ovfl :0          |
-----diagnosis-----
/mnt/610/Hi3516CV610_P0_V1.0.1.0 #
```

检查vi中是否有帧率。

```
-----v1 pipe user pic attr-----
pipe_id enable width height stride pixel_format pool_id phy_addr
0 N 0 0 0 n/a 0 0

-----v1 pipe wrap status-----
pipe_id wrap_en buf_line pts_gap max_gap0 max_gap1 valid_line wrap_err pre_err
0 N 0 n/a n/a n/a n/a n/a n/a

-----v1 pipe status-----
pipe_id enable int_en width height int_cnt send_cnt lost_cnt vb_fail_cnt frame_rate
0 Y Y 2560 1440 1399 1398 0 0 30

-----v1 pipe offline task statistics-----
pipe_id receive_pic_cnt busy_num task_submit_cnt task_fail_cnt task_cost_time max_cost_time
0 1398 0 1398 0 15217 15413

-----v1 phys chn attr1-----
pipe_id chn_id width height pixel_format dynamic_range video_format compress_mode mirror flip
0 0 2560 1440 yvu-sp420 sdr8 linear none N N

-----v1 phys chn attr2-----
pipe_id chn_id depth src_rate dst_rate vb_src vb_pool align
0 0 0 -1 -1 common -1 0

-----v1 phys chn rotation attr-----
pipe_id chn_id rotation type enable rotation
0 0 FIXED N 0

-----v1 ext chn attr1-----
pipe_id chn_id bind_chn src_type width height pixel_format dynamic_range video_format compress_mode

-----v1 ext chn attr2-----
pipe_id chn_id depth src_rate dst_rate vb_src vb_pool align

-----v1 chn low delay attr-----
pipe_id chn_id enable line_cnt one_buf
0 0 N 0 N

-----v1 chn crop info-----
pipe_id chn_id crop_en coord x y width height trim_x trim_y trim_width trim_height
0 0 N ABS 0 0 0 0 0 0 0 0

-----v1 chn status-----
pipe_id chn_id enable width height send_cnt vb_fail_cnt lost_frame_cnt frame_rate
0 0 Y 2560 1440 1398 0 0 30

-----v1 chn out frame info-----
pipe_id chn_id width height compress_mode low_delay_frame delay
0 0 2560 1440 none N 49202

-----v1 interrupt cost time statistics-----
isp_time isp_max_time vicap_time vicap_max_time viproc_time viproc_max_time int_time_per_sec load_ratio_0
18 337 24 892 329 360 32028 44%
/mnt/610/Hi3516CV610_P0_V1.0.1.0 #
```

3 Sensor 驱动 AE 参数配置

完成第二章的内容后，sensor已经具备了基本的出图能力，接下来需要适配ae参数，其中包括增益、曝光、帧率控制三个方面。

3.1 修改曝光、增益、帧率寄存器地址

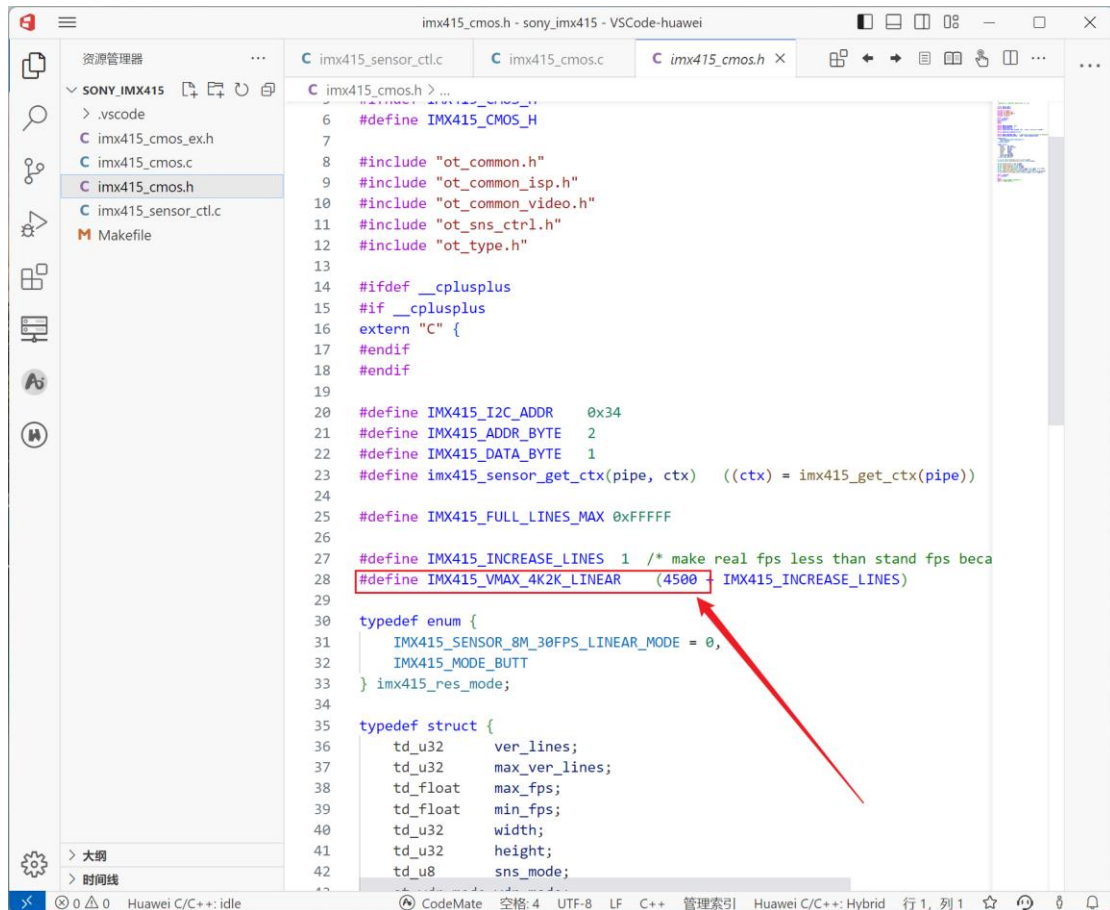
```
84     b1c_clamp_into[vi_pipe] = clamp_en;
85 }
86
87 const imx415_video_mode_tbl g_imx415_mode_tbl[IMX415_MODE_BUTT] = {
88     {IMX415_VMAX_4K2K_LINEAR, IMX415_FULL_LINES_MAX, 30, 0.07,
89      3840, 2160, 0, OT_WDR_MODE_NONE, "IMX415_SENSOR_8M_30FPS_LINEAR"},
90 };
91
92 /*****
93  * local variables
94  *****/
95
96 /* Imx415 Register Address */
97 #define IMX415_VMAX_ADDR_L 0x3024
98 #define IMX415_VMAX_ADDR_M 0x3025
99 #define IMX415_VMAX_ADDR_H 0x3026
100
101 #define IMX415_HMAX_ADDR_L 0x3028
102 #define IMX415_HMAX_ADDR_H 0x3029
103
104 #define IMX415_SHR0_L 0x3050
105 #define IMX415_SHR0_M 0x3051
106 #define IMX415_SHR0_H 0x3052
107
108 #define IMX415_GAIN_PGC_L 0x3090
109 #define IMX415_GAIN_PGC_H 0x3091
110
111 // sensor gain
112 #define IMX415_AGAIN_MIN 1024
113 #define IMX415_AGAIN_MAX 32381
114
115 #define IMX415_DGAIN_MIN 1024
116 #define IMX415_DGAIN_MAX 128914
117
118 #define IMX415_AD_GAIN_TBL_RANGE 241
119 #define IMX415_AGAIN_TBL_RANGE 100
120 #define IMX415_DGAIN_TBL_RANGE 140
121
122 #define IMX415_EXP_TIME_LMT 8
```

3.2 AE 相关参数

检查最大模拟增益，最小模拟增益，最大数字增益，最小数字增益，曝光时间的offset。

```
98 #define IMX415_VMAX_ADDR_H 0x3026
99
100 #define IMX415_HMAX_ADDR_L 0x3028
101 #define IMX415_HMAX_ADDR_H 0x3029
102
103 #define IMX415_SHR0_L 0x3050
104 #define IMX415_SHR0_M 0x3051
105 #define IMX415_SHR0_H 0x3052
106
107 #define IMX415_GAIN_PGC_L 0x3090
108 #define IMX415_GAIN_PGC_H 0x3091
109
110 // sensor gain
111 #define IMX415_Again_MIN 1024
112 #define IMX415_Again_MAX 32381
113
114 #define IMX415_DGain_MIN 1024
115 #define IMX415_DGain_MAX 128914
116
117 #define IMX415_AD_GAIN_TBL_RANGE 241
118 #define IMX415_Again_TBL_RANGE 100
119 #define IMX415_DGain_TBL_RANGE 140
120
121 #define IMX415_EXP_TIME_LMT 8
122
123
124 #define imx415_err_mode_print(sns_image_mode, sns_state) \
125 do { \
126     isp_err_trace("Not support! Width:%u, Height:%u, Fps:%f, WDRMode:%u", \
127                 (sns_image_mode)->width, \
128                 (sns_image_mode)->height, \
129                 (sns_image_mode)->fps, \
130                 (sns_state)->wdr_mode); \
131 } while (0)
132
133 static void cmos_get_ae_comm_default(ot_vi_pipe vi_pipe, ot_isp_ae_sens
134 const ot_isp_sns_state *sns_state)
135 {
```

3.3 修改 vmax 的值



```
6  #define IMX415_CMOS_H
7
8  #include "ot_common.h"
9  #include "ot_common_isp.h"
10 #include "ot_common_video.h"
11 #include "ot_sns_ctrl.h"
12 #include "ot_type.h"
13
14 #ifdef __cplusplus
15 #if __cplusplus
16 extern "C" {
17 #endif
18 #endif
19
20 #define IMX415_I2C_ADDR 0x34
21 #define IMX415_ADDR_BYTE 2
22 #define IMX415_DATA_BYTE 1
23 #define imx415_sensor_get_ctx(pipe, ctx) ((ctx) = imx415_get_ctx(pipe))
24
25 #define IMX415_FULL_LINES_MAX 0xFFFFF
26
27 #define IMX415_INCREASE_LINES 1 /* make real fps less than stand fps beca
28 #define IMX415_VMAX_4K2K_LINEAR (4500 + IMX415_INCREASE_LINES)
29
30 typedef enum {
31     IMX415_SENSOR_8M_30FPS_LINEAR_MODE = 0,
32     IMX415_MODE_BUTT
33 } imx415_res_mode;
34
35 typedef struct {
36     td_u32 ver_lines;
37     td_u32 max_ver_lines;
38     td_float max_fps;
39     td_float min_fps;
40     td_u32 width;
41     td_u32 height;
42     td_u8 sns_mode;
```

vmax_val的计算方式：例如某sensor控制帧率率的vmax寄存器为0x3021 0x3022，其初始化序列中设置的初始值为：

0x3021（低位）：0x22

0x3022（高位）：0x06

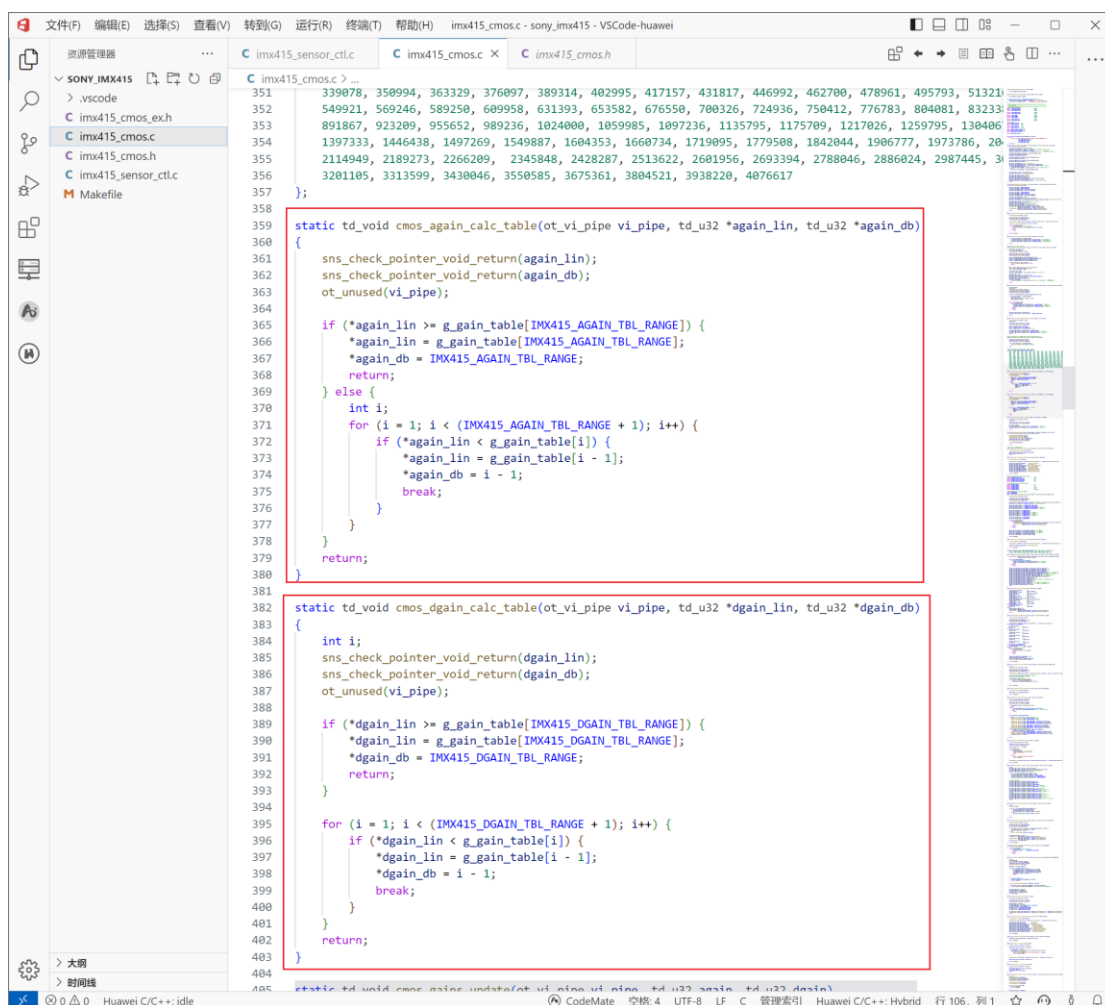
则vmax的值为 $0x0622 = 1570$

3.4 模拟增益、数字增益

检查这个表格是否正确。

```
328 imx415_sensor_get_ctx(vi_pipe, sns_state);
329 sns_check_pointer_void_return(sns_state);
330
331 if (sns_state->wdr_mode == OT_MDR_MODE_NONE) {
332     cms_inttime_update_linear(vi_pipe, int_time);
333 }
334 return;
335 }
336
337 /* Again and Dgain use the same table, Step is 0.3db */
338 static td_u32 g_gain_table[IMX415_AD_GAIN_TBL_RANGE] = {
339     1024, 1059, 1097, 1135, 1175, 1217, 1259, 1304, 1349, 1397, 1446, 1497, 1549, 1604,
340     1660, 1719, 1779, 1842, 1906, 1973, 2043, 2114, 2189, 2266, 2345, 2428, 2513, 2601,
341     2693, 2788, 2886, 2987, 3092, 3201, 3313, 3430, 3550, 3675, 3804, 3938, 4076, 4219,
342     4368, 4521, 4680, 4845, 5015, 5191, 5374, 5562, 5758, 5960, 6170, 6387, 6611, 6843,
343     7084, 7333, 7591, 7857, 8133, 8419, 8715, 9021, 9338, 9667, 10006, 10358, 10722, 11099,
344     11489, 11893, 12311, 12743, 13191, 13655, 14135, 14631, 15146, 15678, 16229, 16799, 17390, 18001,
345     18633, 19288, 19966, 20668, 21394, 22146, 22924, 23730, 24564, 25427, 26320, 27245, 28203, 29194,
346     30220, 31282, 32381, 41519, 34697, 35917, 37179, 38485, 39838, 41238, 42687, 44187, 45740, 47347,
347     49011, 50734, 52517, 54362, 56273, 58250, 60297, 62416, 64610, 66880, 69230, 71663, 74182, 76789,
348     79487, 82281, 85172, 88165, 91264, 94471, 97791, 101228, 104785, 108468, 112279, 116225, 120310, 124537,
349     128914, 133444, 138134, 142988, 148013, 153215, 158599, 164172, 169942, 175914, 182096, 188495, 195119, 201976,
350     209074, 216421, 224027, 231900, 240049, 248485, 257217, 266256, 275613, 285299, 295325, 305703, 316446, 327567,
351     339078, 350994, 363329, 376097, 389314, 402995, 417157, 431817, 446992, 462700, 478961, 495793, 513216, 531251,
352     549921, 569246, 589250, 609958, 631393, 653582, 676550, 700326, 724936, 750412, 776783, 804081, 832338, 861589,
353     891867, 923209, 955652, 989236, 1024000, 1059985, 1097236, 1135795, 1175709, 1217026, 1259795, 1304067, 1349895,
354     1397333, 1446438, 1497269, 1549887, 1604353, 1660734, 1719095, 1779508, 1842044, 1906777, 1973786, 2043149,
355     2114949, 2189273, 2266209, 2345848, 2428287, 2513622, 2601956, 2693394, 2788046, 2886024, 2987445, 3092431,
356     3201105, 3313599, 3430046, 3550585, 3675361, 3804521, 3938220, 4076617
357 };
358
359 static td_void cms_again_calc_table(ot_vi_pipe vi_pipe, td_u32 *again_lin, td_u32 *again_db)
360 {
361     sns_check_pointer_void_return(again_lin);
362     sns_check_pointer_void_return(again_db);
363     ot_unused(vi_pipe);
364
365     if (*again_lin >= g_gain_table[IMX415_AGAIN_TBL_RANGE]) {
```

(2) 检查计算的gain是否正确。again_lin, again_db, dgain_lin, dgain_db 既是输入参数，也是输出参数。当作输入时，去匹配gain table中的值，修正或换算后，当作输出，传入cms_gains_update(ot_vi_pipe vi_pipe, td_u32 again, td_u32 dgain)函数设置寄存器的值。由于gain的计算涉及对数运算，比较消耗算力，因此使用查表法。Gain table可以找sensor原厂获取。



3.5 等曝光量测试

此时可以开始做等曝光量测试，来验证sensor的AE是否配置成功。参考sensor驱动验证文档进行测试。

3.6 aiisp 使用注意

注意使用AIISP时要额外注意sensor dpc强烈建议关闭（可能由于dpc改变原始噪声形态出现pattern）、使用AIISP时建议将dg限制在1024（高增益下dg会导致blc漂移）使用ispdg代替。

4 参数标定

完成参数标定，涉及到：AWB(自动白平衡)、BLC (背光补偿)、CCM (色彩

校正矩阵)、NR (降噪)、LDC镜头畸变校正。如遇到问题或者没有没有实验环境，可以提单协助解决